

Project Manager Maneuver Ammunition Systems
Product Lead Special Ammunition and Weapon Systems
GENERAL SPECIFICATION FOR
ROCKET PROPELLED & SPIN STABILIZED
GRENADE AMMUNITION
10 February 2026

1. SCOPE

1.1 Scope. This general specification describes the performance requirements and verification procedures for the Rocket Propelled and Spin Stabilized Grenade Ammunition.

2. APPLICABLE DOCUMENTS

2.1 General. In addition to documents listed in this section, there are documents specified in section 5 of this specification. All documents listed in this specification should use the latest revision available at the time of award.

2.2 Government documents. Not Applicable.

2.3 Non-Government publications.

- a) United Nations Recommendations on the Transport of Dangerous Goods Model Regulations, 24th Edition, 2025
- b) International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air, 2025-2026 Edition.
- c) International Maritime Organization (IMO) International Maritime Dangerous Goods Code (IMDG) 2024 Edition, Amendment 42-24.

2.4 Acquisition requirements. Acquisition documentation (base award and/or Request for Proposal (RFP)) must specify the following:

- a. Title and date of this specification.
- b. Requirements for certificates of conformance for each lot or shipment of product.
- c. Requirements for age of ammunition.
- d. Requirements and provisions for submission of data as required.
- e. Requirements for acceptance criteria if different than those stated in specification.
- f. Requirements for reduction or elimination of verification procedures.
- g. Requirements and provisions for contractor and Government verification.
- h. Requirements and provisions for packaging of ammunition.
- i. Requirements and provisions for transportation of ammunition.

2.5 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Requirements specified in the contract or delivery order precede the text of this document. Nothing in this document or the contract, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Technical requirements. Ammunition shall meet the requirements and verifications of this specification for the rocket propelled and spin stabilized grenade ammunition listed in ANNEX A. Any grenade ammunition that can be categorized as rocket propelled or spin stabilized but is not specifically identified in ANNEX A must still adhere to the testing parameters of this specification. All rocket propelled and spin stabilized grenade ammunition shall be accompanied with technical data (see APPENDIX) that meets or exceeds the requirements of this specification.

3.1.1 Compliance to technical requirements and storage history records. A Technical Data Package (TDP) as well as lot history folders shall be provided to ensure ammunition is in compliance with this specification. Additionally, documented storage history of the ammunition shall be provided.

3.2 Conformance Testing and Inspection. A random sample of the ammunition/items from each lot presented shall be selected by a U.S. Government (USG) Representative and subjected to conformance testing and inspection in accordance with TABLE I (to include retest rounds, if required).

3.2.1 Stock ammunition. All grenade ammunition must be less than 5 years old and meet the verification procedures in TABLE I of this specification. Any lot that has not been adequately tested to comply with all requirements of Table I shall be subjected to conformance testing and inspection in accordance with this specification.

3.3. Serviceability. Grenade ammunition shall be serviceable and issuable, in good condition and without visible signs of degradation. All parts and assemblies shall be free of burrs, folds, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion, and other foreign material that may adversely affect fit or function. Coatings and protective finishes shall be uniform and shall provide complete coverage of designated surfaces. The painting and marking shall be restored by the assembling facility on any components that have been marred or obliterated.

3.3.1 Identification of defects. Visual defects shall be identified within the TDP and include a defined method for acceptance/rejection of the grenade ammunition. Visual inspections listed in ANNEX C shall be the minimum requirement.

3.3.2 Weight zones. A table outlining weight ranges by zone is required in the TDP if there are multiple projectile weight zones.

3.4 Weapon interface and interoperability. The grenade ammunition shall be functionally compatible with the weapons listed in ANNEX A or as otherwise specified in the contract or delivery order.

3.5 Operating requirements. The grenade ammunition shall provide their functional, operational, and performance capabilities when fired from the weapons listed in ANNEX A, regardless of weather related conditions (example: rain, snow, mud, wind, fog). If testing occurs in inclement weather, the

Contractor shall implement all necessary safety precautions.

3.5.1 Operating temperatures. The grenade ammunition shall demonstrate safety and reliability throughout exposure to ambient and extreme temperatures. All cartridges are required to reliably perform within a temperature range of -30°C to +50°C.

3.5.2 Muzzle velocity. The average velocity of the grenade ammunition obtained at ambient temperature shall meet the requirements listed in ANNEX B. Each lot shall meet the standard deviation requirements. The value of the deviation if not provided in Request for Proposal (RFP) will be determined during TDP development.

3.5.3 Chamber pressure. The individual max pressure of the grenade ammunition obtained at ambient shall meet the requirements listed in ANNEX B. The average pressure at ambient and the individual maximum pressure obtained at hot shall meet the requirements in the TDP.

3.5.4 Maximum effective range. The maximum effective range of the grenade ammunition when tested at ambient shall meet or exceed the requirements listed in ANNEX B.

3.5.5 Dispersion. The grenade ammunition shall demonstrate dispersion characteristics at ambient as specified in the TDP.

3.5.6 Function and casualty. The grenade ammunition shall function without casualty or cause damage to the weapon, and shall exhibit a ballistic performance free of misfires, duds, premature bursts, short rounds, and metal parts separation. The ammunition shall cause no hazard or unsafe conditions to the user during storage, transportation, or use.

3.5.7 Fragmentation. The grenade ammunition shall provide the lethal fragmentation at the desired radius for each item as specified in ANNEX B. (High Explosive/Fragmentation (HE/FRAG) Grenades only).

3.5.8 Penetration. The grenade ammunition shall meet the armor penetration characteristics as specified in ANNEX B. (High Explosive Anti-Tank (HEAT) Grenades only).

3.6 Fuze identification and description. Within each TDP the fuze shall be identified with a drawing/figure of the internal fuze design with components categorized to include the type and weight of energetic material(s) and a functional description of the fuze that includes, at a minimum, a detailed description of the fuze arming sequence, operating temperatures, in-process tests (example: jolt/jumble, sensitivity, safe arming/non-arming etc.), and no arm and all arm distances. Test records shall be available to the USG for verification. Fuze shall be set to safest setting, if applicable.

3.7 Environmental Requirements.

3.7.1 Sequential rough handling. Grenade ammunition shall be capable of withstanding the rigors of the sequential rough handling and transportation throughout extreme temperature ranges and meet all performance and safety requirements.

3.7.2 Shelf life. The grenade ammunition shall meet all requirements of this specification within its operating temperature range at a minimum of ten (10) years. All cartridges are required to reliably be stored within a temperature range of -30°C to +50°C at a minimum of ten (10) years.

4. VERIFICATION

4.1 Technical verification. The grenade ammunition shall be subjected to verification of all requirements cited in Section 3 in this specification, in accordance with the TABLE I. Noncompliance with any requirements shall be cause to withhold acceptance of the lot or batch in which the noncompliance was found.

TABLE I. Requirements/Verification Cross-Reference Matrix

Section 3 Requirements	Requirement	Section 4 Verification Procedures	Sample Size
3.1	Technical requirements	4.1	-
3.3	Serviceability	4.3	Note 1
3.3.1	Identification of defects	4.3	ANNEX C
3.4	Weapon interface and interoperability	4.4	ANNEX A
3.5.1	Operating temperatures	4.5.1	ANNEX A
3.5.2	Muzzle velocity	4.5.2	ANNEX B
3.5.3	Chamber Pressure	4.5.3	ANNEX B
3.5.4	Maximum effective range	4.5.4	ANNEX B
3.5.5	Dispersion	4.5.5	ANNEX B
3.5.6	Function and casualty	4.5.6	ANNEX A
3.5.7	Fragmentation	4.5.7	ANNEX B
3.5.8	Penetration	4.5.7	ANNEX B
5.1	Packaging	-	ANNEX C
5.3	Packing	-	ANNEX C
5.5	Marking	-	ANNEX C
Note 1: To be performed using a defined sampling procedure for inspection determined by the ammunition lot size(s).			

4.1.1 Lot formation. The grenade ammunition shall be assembled into identifiable lots, sublots, or batches, or in such other manner contingent on approval by the USG. Each lot or batch shall, as far as practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time. Lot quantities shall be determined by the total quantity for shipment and include the additional rounds from the same lot to be consumed during lot acceptance testing along with contingency rounds for potential retesting. The lots or batches shall be identified by the contractor and shall be kept intact in adequate and suitable storage space. All grenades in a given lot must be manufactured prior to lot acceptance testing.

4.1.1.1. Lot Size. All lot sizes are to remain consistent throughout an awarded delivery order. Once temperature testing is conducted, all skip lot sizes shall be within 20% of that lot's size. If a lot size larger than 20% is proposed, temperature testing has to be conducted on that lot. If a lot fails for any reason, the next two lots presented for additional testing have to be within 20% of the failed lot size.

4.1.1.2 Component Lots. If multiple lots of a component are present, a minimum of 10 rounds shall be fired for each lot.

4.1.2 Retest criteria. A ballistic retest is authorized upon initial failure of ammunition to meet test requirements as long as the failure is not considered critical. Retest procedures must be clearly defined in the TDP and results must be combined with data from the first test unless a failure analysis confirms that the failure was caused by something other than the ammunition (example: weapon failure or faulty test set up).

4.2 Verification of compliance. Evidence shall include, but is not limited to, the following:

- a. Identification of technical requirements to which the grenades is/was produced.
- b. Manufacturer, date of manufacture, and original acceptance documentation.
- c. Initial acceptance reports, including type, lot or batch identification, quantity and method of acceptance (example: sample size, verification method, acceptance criteria, results...).
- d. Surveillance reports, including lot or batch identification, quantity, and method of surveillance (example: sample size, verification method, criteria for action to be taken on lot or batch, results).
- e. Storage history, including duration, and storage condition (example: controlled, uncontrolled).

4.3 Serviceability and identification of defects. A random sample shall be selected from each lot of grenades for inspection. Grenades and packaging shall be of good condition without visible signs of degradation. The grenades shall be visually inspected in accordance with ANNEX C of this specification. Reinspection of a lot that fails for non-ballistic reasons is authorized as long as the criteria for reinspection is clearly outlined in the technical data.

4.3.1 Weight zone verification. For items with multiple projectile weight zones, 24 samples must be weighed and fall within the allotted ranges in the weight zone table provided in the TDP.

4.4 Weapon interface and interoperability. A random sample shall be selected from each lot of grenades to be delivered and inserted into a weapon chamber or chamber gage that conforms to the identified weapon system to check for profile and alignment. Inability to interface properly with weapon system shall be considered a failure to show compliance and constitute a single reliability failure.

4.5 Operating verification. The same cartridge of ammunition can be used for multiple tests provided the test equipment is capable of capturing all necessary test data. For example; Muzzle Velocity, Maximum Effective Range, and Function & Causality testing can be done at the same time using the same cartridges. Called warmers and spotters are allowed and encouraged. “Called” means that before the cartridge is loaded into the weapon, the USG representative must be informed that the next cartridge to be fired is a warmer or spotter. Should a cartridge not be “Called” it shall be counted toward whichever test is being conducted, potentially resulting in a LAT failure. Safety failures apply to all cartridges fired during LAT, regardless of which test is being performed. The following safety failures shall result in the immediate rejection of the lot; there shall be no retest:

- a. Detonation, Deflagration, or function within the weapon (all cartridge types).

- b. Detonation of an High-Explosive (HE) or High-Explosive Anti-Tank (HEAT) cartridge before impact with the ground or target.
- c. An early function of a Practice (Live) cartridge before impact with the ground or target which exposes the operator to smoke, fragmentation or fire.
- d. An early function of Smoke, Airburst, or Illumination cartridge which exposes the operator to smoke, fragmentation or fire.
- e. Metal parts separation.
- f. A cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools).

4.5.1 Operating temperatures. Grenade ammunition shall operate reliably and safely when conditioned and fired throughout a reasonable extreme operating temperature range. Ammunition shall consistently perform within a temperature range of -30°C to +50°C, ensuring reliable function under the specified environmental conditions. Temperature testing shall be conducted on a skip-lot basis (see APPENDIX) during LAT. A single safety failure will result in lot rejection and there shall be no retest. Functional testing parameters must be defined in the technical data and meet manufacturer requirements. The entire cartridge, including all components necessary to fire and function (example: fuze, warhead, propellant, etc.) shall be temperature conditioned based on the maximum cartridge body diameter. Additionally, the cartridges shall be fired within the following times after being removed from the conditioning chamber or thermally insulated bag.

- a. Ammunition less than 85mm in diameter shall be conditioned for not less than six (6) hours and fired within five (5) minutes of removal from conditioning box or thermally insulated bag.
- b. Ammunition between 85mm and 120mm in diameter shall be conditioned for not less than nine (9) hours and fired within six (6) minutes of removal from conditioning box or thermally insulated bag.
- c. Ammunition greater than 120mm in diameter shall be conditioned for not less than twelve (12) hours and fired within eight (8) minutes of removal from conditioning box or thermally insulated bag.

4.5.1.1 Projectile weight and charge operational test criteria: If there are multiple projectile weight zones, the lowest weight zone available shall be conditioned and tested at the lowest operating temperature and the highest weight zone available conditioned and tested at the highest operating temperature. The highest weight zone will also be utilized for ambient testing. In addition, the minimum propellant charge zone shall be used at the lowest operating temperature, and the maximum propellant charge zone shall be used at the highest operating temperature.

4.5.2 Muzzle velocity. Velocity tests shall be conducted in accordance with the manufacturer's test procedures to verify that a sample representative of the lot meets the velocity requirements specified in ANNEX B (10 cartridges minimum). If the lot fails to meet the Velocity requirements a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum) shall be used to determine if the lot passes. Each lot shall meet the standard deviation requirements specified in the RFP or TDP.

4.5.3 Chamber pressure. Pressure tests shall be conducted in accordance with the manufacturer's test procedures to verify samples from the lot do not exceed the maximum individual pressure requirements listed in ANNEX B (10 cartridges minimum), the average pressure at ambient listed in the TDP (10 cartridges minimum), and the maximum individual pressure requirements at hot

listed in the TDP (10 cartridges minimum). If the lot fails to meet the Pressure requirements a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum) shall be used to determine if the lot passes. Any single round that exceeds the maximum individual pressure requirement will result in lot rejection, there shall be no retest.

4.5.4 Maximum Effective Range. Maximum effective range shall be verified during lot acceptance testing to ensure compliance with ANNEX B.

4.5.5 Dispersion. Dispersion tests shall be conducted to verify that each lot of ammunition meets the specified requirements in the TDP (10 cartridges minimum). If the lot fails to meet the Dispersion requirements or a round misses the target, a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum or 19 cartridges if one misses the target), shall be used to determine if the lot passes. If two (2) or more cartridges miss the target (cumulative) the lot fails.

4.5.6 Function and casualty. A random sample shall be selected from the ammunition lot in accordance with ANNEX A to be delivered and functioned from the identified weapon system to check for function, casualty and metal parts security. The ammunition shall function across the entire operating temperature range without casualty or damage to the weapon, and shall exhibit a ballistic performance free of misfires, duds, premature bursts, short rounds, and metal parts separation. A short round is defined as a cartridge that has half of the muzzle velocity of the others within its group and/or impacts the ground at a range 50% or less of the range average or its respective temperature/charge/group; excluding the cartridge(s) in question. If a cartridge has a multi-option fuze, all options shall be tested during Lot Acceptance Testing to properly demonstrate the fullest extent of the munitions capability unless specified otherwise in Annex A or by the USG during solicitation. The different fuze options test quantity will be distributed amongst the total Function and Casualty requirement. The ammunition shall cause no hazard or unsafe conditions to the user during storage, transportation or use. Critical safety failures are listed in Section 4.5 of this specification.

4.5.7 Performance. Proper functioning for each grenade ammunition type is defined below. Any round which does not function properly according to the definitions below, the requirements in ANNEX A and B, and the requirements in the TDP are considered a function failure:

- a. Fragmentation: The ammunition, with a point detonate fuze, shall produce a high-order detonation of the warhead upon initial impact with the target (terrain or man-made structure) as specified in the TDP. If fuze contains a delay the ammunition shall detonate after impact with target at a specific time or distance as specified in the TDP.
- b. Bounding: The ammunition shall bound/jump after initial impacting with the ground and produce a high-order detonation of the warhead above ground between a low/high height as specified by the manufacturer in the TDP.
- c. Airburst: The ammunition shall produce a high-order detonation of the warhead above ground between a min/max distance as specified by the manufacturer in the TDP.
- d. Proximity: The ammunition shall produce a high-order detonation of the warhead within a min/max distance of target as specified by the manufacturer in the TDP.
- e. HEAT: The ammunition shall produce a high-order detonation resulting in jet capable of penetrating armor at a depth as specified in the TDP.
- f. Smoke: The ammunition shall produce a volume of smoke upon impact with the target as specified by the manufacturer in the TDP.

- g. Illumination: The ammunition shall produce parachute suspended light at specific intensity, duration, wavelength, and height as specified by the manufacturer in the TDP.
- h. Penetration: The ammunition shall meet the armor penetration characteristics as specified in the TDP.
- i. Practice (Inert): The ammunition shall not contain any explosives or energetics other than propellant.
- j. Practice (Live): The ammunition may contain energetics, other than propellant, to signal location of impact.

5. PACKAGING

5.1 Preservation, packaging, packing, unitization, and marking shall provide protection for multiple handling, redistribution, and shipment by any transportation mode and meet or exceed the following requirements.

5.1.1 Packaging containers for hazardous materials, ammunition and explosives shall meet or exceed the requirements found in part 6 of the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations" and in a manner acceptable to the competent authority of the nation of origin and in accordance with the regulations of all applicable carriers.

5.1.2 Cleanliness – Items and packaging shall be free of dirt and other contaminants which would contribute to the deterioration of the item or which would require cleaning by the customer prior to use. Coatings and preservatives applied to the item for protection are not considered contaminants.

5.1.3 Preservation – Items susceptible to corrosion or deterioration shall be provided protection against external environmental effects.

5.1.4 Cushioning – Items requiring protection from physical and mechanical damage (example: fragile, sensitive, critical material) or which could cause physical damage to other items, shall be protected by wrapping, cushioning, pack compartmentalization, or other means to mitigate shock and vibration and prevent damage during handling and shipment.

5.1.5 Wood Packing Material (WPM) – All natural wood packing material shall be constructed from heat treated lumber and certified by an accredited recognized agency in accordance with International Standards for Phytosanitary Measures (ISPM-15) regardless of origin or destination. All natural wood packing materials must include certification markings that are readily visible to inspectors and on at least two sides of every container when palletized.

5.1.6 Firing Tables – A copy of the manufacturer's firing tables for ammunition shall be included in the TDP.

5.2. Unit Package

5.2.1 Unit package shall be so designed and constructed that it will contain the contents with no damage to the item(s), with minimal damage to the unit pack during shipment and storage in the shipping container and will allow subsequent handling. The outermost component of the unit package shall be a container such as a sealed bag, carton, or box.

5.3. Packing

5.3.1 Unit packages must be packed in shipping containers. All shipping containers shall be the most cost effective and shall be of the minimum cube to contain and protect the items.

5.3.2 Shipping Containers – The shipping container (including any necessary blocking, bracing, cushioning, or waterproofing) shall comply with the regulations of the carrier used and shall provide safe delivery to the destination at the lowest tariff cost. The shipping container shall be capable of multiple handling, stacking at least ten feet high, and storage under favorable conditions and meet the requirements of the "United Nations Recommendations on the Transport of Dangerous Goods".

5.4. Unitization

5.4.1 Shipments of identical items going to the same destination shall be palletized if they have a total cubic displacement of 20 cubic feet or more unless skids or other forklift handling features are included on the containers. Pallet loads must be stable, and to the greatest extent possible, provide a level top for ease of stacking. The weight capacity of the pallet must be adequate for the load. A pallet load shall not exceed 4,000 pounds and should not exceed 52 inches in length or width, or 54 inches in height. The load shall be contained in a manner that will permit safe handling during shipment and storage.

5.4.2 Banding - Metal banding shall be used to secure the load. Straps shall be applied to each column or layer of boxes. Tie down straps shall be applied to each column of boxes at 90 degrees to the load straps. Edge protectors shall be used when securing fiberboard boxes.

5.5. Marking

5.5.1 Packaging marking shall be visible, clear, and remain legible during normal life cycle handling.

5.5.2 All unit packages, intermediate packs, exterior shipping containers, and as applicable, unitized loads shall be marked with item description, quantity, lot number, or serial number. The outer shipping container and unitized load shall indicate load weight, UN dangerous goods proper shipping name, and location of UN number and UN POP marking.

5.5.3 Each package shipping container shall show the United Nations packaging symbol and applicable codes in accordance with the construction requirements and testing of packaging as expressed in part 6 of the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations".

5.6. Additional Requirements for Hazardous Materials

5.6.1 The shipment shall be prepared in accordance with the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations" and other applicable regulations effective at the time of shipment in a manner acceptable to the competent authority of the nation of origin and in accordance with the regulations of all applicable carriers.

5.6.2 Packaging and marking for hazardous material shall comply with the requirements for the mode of transport and the applicable performance packaging contained in the following documents:

- a. International air transport: International Civil Aviation Organization (ICAO)

Technical Instructions for the Safe Transport of Dangerous Goods by Air.

- b. International vessel transport: International Maritime Organization (IMO)
International Maritime Dangerous Goods Code (IMDG).

6. Reporting

6.1 Lot Acceptance Test Reports (LATR), at a minimum, must include a summary of findings for all visual and ballistic testing, lot history folders (in-process testing per TDP requirements, ammo data card, and all certifications or certificates), lot hall measured data (including weights), pressure (at ambient, hot) and velocity for each individual shot, range for all test shots, deflection for all test shots scored for dispersion, identify conditioning temperature used, calculated average pressure and velocity, spread for minimum and maximum velocity, and calculated dispersion. The LATR must also include the ammo data card for all lots tested.

6.1.1 Ammo Data Cards are required to be provided as part of the LATR and must include lot information for all energetics and major components used in the production of each lot.

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Annex A: Weapon Interface

Annex B: Operating Requirements

Annex C: Visual Inspection Requirements

APPENDIX – Definitions

ANNEX A - Weapon Interface

Spin Stabilized Grenades					
Item ^{1,8,9}	Weapon Interface ¹⁰	Function and Casualty/Safety Order size greater than 249 cartridges Sample Size ² -Accept-Reject & Retest (cumulative)			Function and Casualty/Safety Order size 100-249 cartridges ³ Sample Size ² -Accept-Reject & Retest (cumulative)
		Ambient	Hot	Cold	Ambient Only
40mm HE Frag. ^{6,7} 40mm HE Frag. Bounding ^{4,7} 40mm HE Frag. Airburst ⁵ 40mm Smoke ⁷ 40mm Illumination 40mm Practice	GP 25/30 Grenade Launcher	26-0-1 Retest 46-1-2 (Cumulative)	26-1-2 Retest 46-2-3 (Cumulative)	26-1-2 Retest 46-2-3 (Cumulative)	26-0-1 Retest 46-1-2 (Cumulative)
		Out of Mode Function (Bounding & Airburst) 26-3-4 Retest 46-5-6 (Cumulative)	Out of Mode Function (Bounding & Airburst) 26-3-4 Retest 46-5-6 (Cumulative)	Out of Mode Function (Bounding & Airburst) 26-3-4 Retest 46-5-6 (Cumulative)	Out of Mode Function (Bounding & Airburst) 26-3-4 Retest 46-5-6 (Cumulative)
Notes on next page					

Note 1: Any spin stabilized grenade ammunition not listed in this table are still required to meet sample size-accept-reject criteria listed in this table.

Note 2: These samples sizes shall be used to show performance and function and casualty at ambient temperature. 26 rounds shall be tested for function and safety at the operating temperature extremes. A minimum of 10 data points is required to evaluate all other test parameters.

Note 3: For orders less than 100 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 4: Bounding cartridges shall jump/bound after first impact and detonate at the intended height. A cartridge that detonates outside of the intended height, but within the lethal radius of the cartridge from initial impact shall be counted as an out of mode function. A cartridge that detonates outside of the lethal radius from initial point of impact shall be counted as a function failure.

Note 5: Airburst cartridges shall detonate above ground at the intended distance. A cartridge that detonates outside of the intended distance, but within the lethal radius of the cartridge from initial impact shall be counted as an out of mode function. A cartridge that detonates outside of the lethal radius from initial point of impact shall be counted as a function failure.

Note 6: For these cartridge types only, if they contain a multi-option fuze where the only options are Point Detonate and Delay, they shall be set to Point Detonate during LAT.

Note 7: If the fuze has an adjustable time delay setting it must be set as per approved TDP during Function & Casualty testing.

Note 8: A function failure shall be defined as:

- a) A cartridge that will not load into the weapon.
- b) A misfire - failure of a projectile to discharge as intended
- c) A short round.
- d) Smoke cartridges only: A cartridge that does not produce specified volume of smoke or has early function that does not expose user to smoke, blast, fire, or fragmentation.
- e) Illumination cartridges only: A cartridge that does not produce light for specified intensity, wavelength, duration and height or has early function that does not expose user to fire, blast, or fragmentation.
- f) HE cartridges only: A cartridge that does not completely detonate upon initial impact (excluding bounding type grenades) with their target when equipped with a point detonate fuze. Self-destruct detonation or fuze function only shall be scored as a function failure, if shot as part of a function test. If fuze is set to delay the ammunition shall detonate after impact with target at a specific time or distance as specified in the TDP.
- g) Practice (Live) cartridges only: A cartridge that has an early function, or a failure to function before impact with the ground or target that does not expose user to smoke, blast, fire, or fragmentation.
- h) A cartridge that does not perform its primary function.

Note 9: Critical safety failures are listed in Section 4.5 of this specification.

Note 10: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved for use by the USG.

ANNEX A - Weapon Interface (continued)

Rocket Propelled Grenades					
Item ^{1,4,5,6,7}	Weapon Interface ⁸	Function and Casualty/Safety Order size greater than 249 cartridges Sample Size ² -Accept-Reject & Retest (cumulative)			Function and Casualty/Safety Order size 100-249 cartridges ³ Sample Size ² -Accept-Reject & Retest (cumulative)
		Ambient	Hot	Cold	Ambient only
40mm HE/Frag Grenade (OG-7V) 40/70.5mm HEAT (PG-7VM) 40/93mm HEAT (PG-7VL) 40mm Practice	RPG-7 Grenade Launcher	32-0-1 Retest 60-1-2 (Cumulative)	32-1-2 Retest 60-2-3 (Cumulative)	32-1-2 Retest 60-2-3 (Cumulative)	32-0-1 Retest 60-1-2 (Cumulative)
73mm HE (OG-9V) 73mm HEAT (PG-9V) 73mm Practice	SPG-9 Recoilless Gun	15-0-1 Retest 29-1-2 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-0-1 Retest 29-1-2 (Cumulative)
Notes on next page					

Note 1: Any rocket propelled grenade ammunition not listed in this table are still required to meet sample size-accept-reject criteria listed in this table for which type of weapon system it most closely resembles.

Note 2: These samples sizes shall be used to show performance and function and casualty at ambient temperature. The rounds shall be tested for function and safety at the operating temperature extremes (32 rounds for RPG-7 ammunition and 15 rounds for SPG-9 ammunition). A minimum of 10 data points is required to evaluate all other test parameters.

Note 3: For orders less than 100 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 4: If the fuze has an adjustable time delay setting it must be set as per approved TDP during Function & Casualty testing.

Note 5: For these cartridge types only, if they contain a multi-option fuze where the only options are Point Detonate and Delay, they shall be set to Point Detonate during LAT.

Note 6: A function failure shall be defined as:

- a) A cartridge that will not load into the weapon.
- b) A misfire - failure of a projectile to discharge as intended
- c) A short round.
- d) HE cartridges only: A cartridge that does not completely detonate upon initial impact with their target. Self-destruct detonation or fuze function shall only be scored as a function failure, if shot as part of a function test.
- e) Practice (Live) cartridges only: A cartridge that has an early function, or a failure to function before impact with the ground or target that does not expose user to smoke, blast, fire, or fragmentation.
- f) A cartridge that does not perform its primary function.
- g) Smoke cartridges only: A cartridge that does not produce specified volume of smoke or has early function that does not expose user to smoke, blast, fire, or fragmentation.
- h) Illumination cartridges only: A cartridge that does not produce light for specified intensity, wavelength, duration and height or has early function that does not expose user to fire, blast, or fragmentation.

Note 7: Critical safety failures are listed in Section 4.5 of this specification.

Note 8: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved for use by the USG.

ANNEX B - Operating Requirements

Spin Stabilized Grenades				
Item ^{3,4}	Average Muzzle Velocity ¹ (m/s)	Max. Individual Pressure ^{1,2} (MPa)	Max. Effective Range ¹ (m)	Lethal Fragmentation Radius ¹ (m)
40mm HE Frag (VOG-25)	≥ 70	≤ 85	≥ 400	≥ 5
40mm HE Frag Bounding (VOG-25P)				
40mm HE Frag Airburst				
40mm Illumination				
40mm GRD-40 Smoke				
Note 1: A minimum of 10 data points shall be required for each test parameter, however, it is permissible to leverage the same rounds to satisfy more than one requirement. Note 2: For Practice cartridges: Pressure, Velocity, and Effective Range shall be per the manufacturer’s requirements unless specified by the USG in the proposals. Note 3: Safety failures are listed in Section 4.5 of this specification.				

ANNEX B - Operating Requirements (continued)

Rocket Propelled Grenades					
Item ^{4,5}	Average Muzzle Velocity ¹ (m/s)	Max. Individual Pressure ^{1,2} (MPa)	Max. Effective Range ¹ (m)	Lethal Frag. Radius ¹ (m)	Armor Penetration ³ (mm)
40mm HE/FRAG (OG-7V)	≥ 145	≤ 95	≥ 950	≥ 15	-
40mm HEAT (PG-7V)	≥ 110		≥ 350	-	≥ 260
40mm HEAT (PG-7VM)	≥ 135		≥ 450	-	≥ 200
40mm HEAT (PG-7VL)	≥ 105		≥ 250	-	≥ 400
73mm HE (OG-9V)	≥ 310	≤ 60	≥ 450	≥ 15	-
73mm HEAT (PG-9V)	≥ 430	≤ 67	≥ 800	-	≥ 300
Note 1: A minimum of 10 data points shall be required for each test parameter, however, it is permissible to leverage the same rounds to satisfy more than one requirement. Note 2: Not less than 80% of the Armor piercing ammunition (10 shot minimum) shall have full penetration of armor. All rounds must be fired with no safety or critical failures. Note 3: For Practice cartridges: Pressure, Velocity, and Effective Range shall be per the manufacturer's requirements unless specified by the USG in the proposals. Note 4: Safety failures are listed in Section 4.5 of this specification.					

ANNEX C – Grenade Visual Inspection Requirements

Critical: 24-0-1	Major: 24-1-2 Reinspection: 72-3-4 (cumulative)	Minor (24-3-4) Reinspection: 72-9-10 (cumulative)
Missing, cracked, loose or bent parts	Cases with moving cartridges; loose pack	Discolored, oily, dirty, or smeared
Gross Corrosion, Pitting, or deformation	Incorrect quantity of cartridges in packaging (light boxes shall have their corners painted) each cartridge missing shall constitute one failure	Minor surface corrosion
Spillage/leakage of energetic materials	Wire Security Seal missing	Dents, chips or burrs
Incorrect Marking of cartridge type: HE cartridge marked as Inert, Practice or Smoke	Mixed ammunition types: Training cartridge mixed with Combat cartridge lot	Scratches on cartridge
Mixed ammunition types: Combat cartridge mixed with training cartridge lot	Missing, illegible or incorrect marking	Excessive grease on cartridge
Bourrelet diameter too large	Incorrect Marking of cartridge type: Any cartridge type other than HE	Irregular disposition of cartridges in case
	Projectile weight is out of specified range	Paint damaged, wet or incomplete
Note 1: If the number of allowable defects is exceeded a double sample quantity shall be re-inspected. The accept/reject criteria is based on the cumulative number of defects for all three samples. Note 2: If a critical is found or the number of allowable majors or minors, after reinspection, is exceeded the lot fails. Note 3: The inspection and reinspection quantity listed is selected during LAT. DCMA will utilize their Zero-Based Acceptance Sample Plans shown in Appendix during GSI.		

APPENDIX

1. DEFINITIONS

1.1 PdL SAWS Rocket Propelled and Spin Stabilized Grenade Ammunition. PdL SAWS rocket propelled and spin stabilized grenade munitions are those munitions that have not been safety tested and type classified for Army use, cannot be procured through the Army supply. Munitions and explosives that are not managed by National Inventory Control Points, have not been safety tested nor type classified for Army use, do not have a national stock number (NSN) and cannot be procured or requisitioned through the Army or other Department of Defense supply system.

1.2 Technical data. Technical data, otherwise known as the Technical Data Package (TDP), is the product specific technical drawings and Quality Assurance (QA) requirements to which ammunition and associated packaging is produced and accepted for each applicable Contract Line Item Number (CLIN). For the purpose of this specification, the TDP, as a minimum, must contain the following: Government Contract Number and Delivery Order; top assembly, fuze and key component drawings with revision number, revision date, document control signature, key interface dimensions, and a list of component assemblies with drawing numbers and revision dates; packaging and marking drawings with revision number, revision date, document control signature and markings; and product specification with final assembly, fuze and key component in-process acceptance test methods with sample size, and accept/reject criteria. Within the TDP there shall also be a clear cut-away and/or exploded drawing(s) to show all of the inner parts and workings of the fuze. Additionally, there shall be paragraphs which describe the fuze arming sequence and events leading to function. A copy of the manufacturer's firing tables for ammunition shall be included with the TDP.

1.3 Degradation. Ammunition with gross nonconformance to identified technical requirements, corrosion, cracks, deformation, and spillage.

1.4 Deterioration. Packaging ripped, broken, perforated, water damaged, or crushed.

1.5 Skip-Lot Temperature Extreme Testing. Testing at the operational temperature limits shall be conducted on the first lot of every delivery order and every 5th lot thereafter (1, 5, 10, 15...). The ammunition must pass the requirements defined in ANNEX A. If a lot is rejected for any reason, the following two lots must pass temperature testing before resuming the skip lot procedures.

APPENDIX (continued)

	Acceptable Quality Level (AQL)															
LOT SIZE	0.01	0.015	0.025	0.04	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10.0
1-8	A	A	A	A	A	A	A	A	A	A	A	A	5	3	3	3
9-15	A	A	A	A	A	A	A	A	A	A	13	8	5	3	3	3
16-25	A	A	A	A	A	A	A	A	A	20	13	8	5	3	3	3
26-50	A	A	A	A	A	A	A	A	32	20	13	8	7	7	5	3
51-90	A	A	A	A	A	A	80	50	32	20	13	13	11	8	5	4
91-150	A	A	A	A	A	125	80	50	32	20	19	19	11	9	6	5
151-280	A	A	A	A	200	125	80	50	32	29	29	19	13	10	7	6
281-500	A	A	A	315	200	125	80	50	48	47	29	21	16	11	9	7
501-1200	A	800	500	315	200	125	80	75	73	47	34	27	19	15	11	8
1201-3200	1250	800	500	315	200	125	120	116	73	53	42	35	23	18	13	9
3201-10,000	1250	800	500	315	200	192	189	116	86	68	50	38	29	22	15	9
10,001-35,000	1250	800	500	315	300	294	189	135	108	77	60	46	35	29	15	9
35,001-150,000	1250	800	500	490	476	294	218	170	123	96	74	56	40	29	15	9
150,001-500,000	1250	800	750	715	476	345	270	200	156	119	90	64	40	29	15	9
500,001 & Over	1250	1200	1112	715	556	435	303	244	189	143	102	64	40	29	15	9

Note: An "A" in place of a number indicates that all units in a lot must be inspected